Label Alignment on Packaging

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Problem Statement

In industrial packaging lines, product labels must be placed correctly before products are shipped to customers. However, manual label inspection has several limitations.

Main Problems

- Human inspectors can become tired during continuous production.
- Manual checking is slower and less consistent.
- Label defects may pass without being detected.
- Incorrect labels can reduce product quality and customer trust.

Common Defects

- Missing Label
- Translation Defect
- Rotation Defect

Therefore, an automated machine vision system is needed to inspect labels accurately, quickly, and consistently.

Project Objectives

01

To Develop

A machine vision system for automatic label alignment inspection.

02

To Detect

Label defects such as missing label, translation defect, and rotation defect.

03

To Classify

Packaging products as Pass or Defective using image processing and decision logic.

Dataset

A small dataset was prepared to simulate label inspection in an industrial packaging line. It covers four critical operational categories to evaluate alignment and correctness.

01

Pass

Label is present, centered, and correctly aligned.

02

Missing Label

Label is not present on the product packaging.

03

Translation

Label is moved from the correct position horizontally or vertically.

04

Rotation Defect

Label is tilted or rotated from the correct angle.

Proposed Methodology

An edge and pattern based pipeline locates each label, then measures how far it deviates from the reference position and angle.

1. **Capture**
   Image the bottle under frontlight
2. **Pre-process**
   Grayscale, denoise, sharpen edges
3. **Detect label**
   Edge / pattern match finds corners
4. **Measure**
   Compute X/Y shift and tilt angle
5. **Classify**
   Apply tolerances, assign state

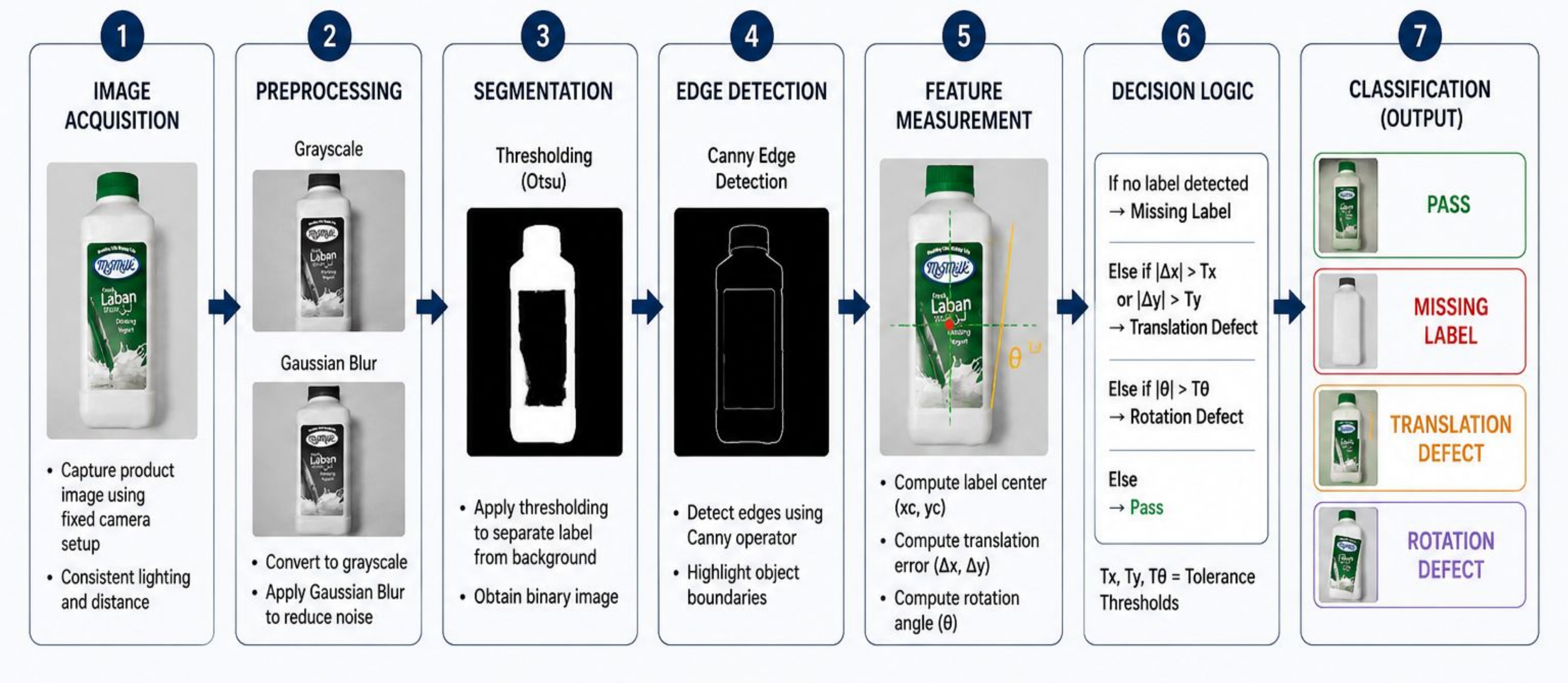
Lighting setup

A directed frontlight or diffused frontlight is chosen based on how glossy the label is, reducing glare so the label edges stay sharp and detectable.

Why it's well-suited

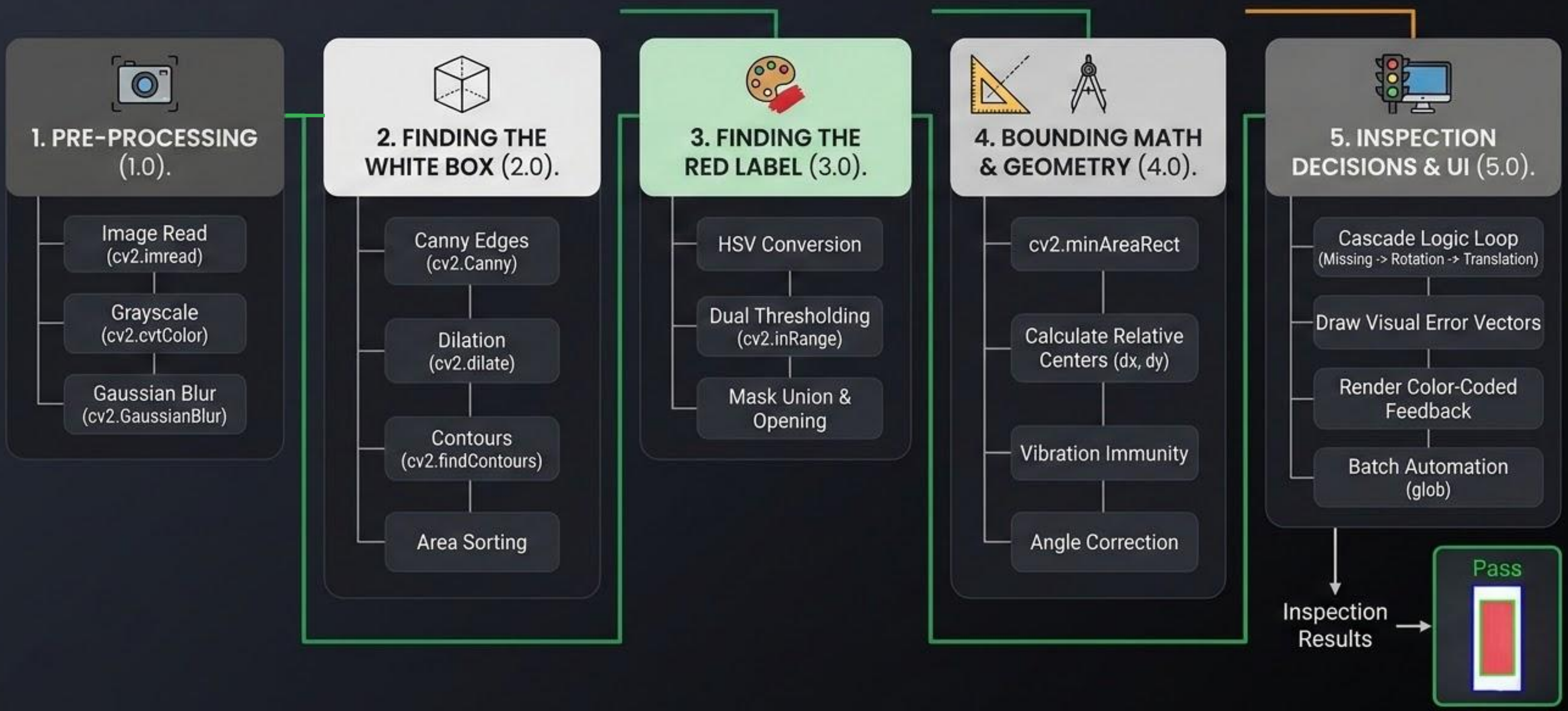
Boxes and bottles are easy to position and image, and label inspection is one of the most common, well-understood factory vision applications.

Proposed Methodology



Data Processing Pipeline for Label Inspection

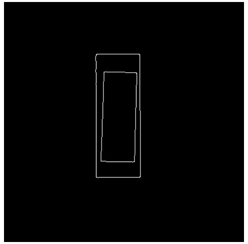
Detailed flow chart of the image analysis and logical sequence.



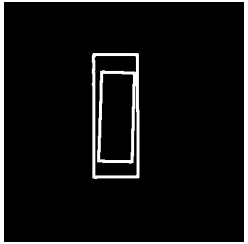
Visual Representation of the Process

1. Raw Image

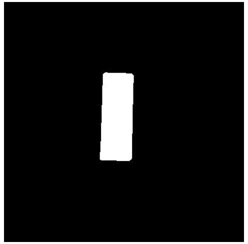
2. Grayscale + Blur



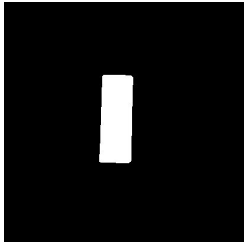
3. Canny Edges



4. Dilated Edges



5. Raw HSV Mask



6. Clean Label Mask

Final Output

Results & Analysis

Detection output & measurement — what the system extracts from each frame.

Illustrative overlay — box (blue), label (green outline), centroids and shift vector. Frames are annotated at runtime by the cv2 script.

1. **Locate package box**
   Largest Canny-edge contour → minAreaRect gives box centre (bx, by).
2. **Locate label**
   Largest red HSV region → minAreaRect gives label centre (lx, ly) and angle θ.
3. **Measure shift**
   $dx = |lx - bx|$, $dy = |ly - by|$, in pixels relative to the box centre.
4. **Normalise tilt**
   θ folded to ±45°, then $|\theta|$ gives the absolute rotation in degrees.
5. **Apply thresholds**
   Compare against 5 px and 1.0° to assign the final verdict.

Results & Analysis

Classification outcomes and analysis of the rule-based inspection.

DECISION RULES

Pass

`dx ≤ 5 px and dy ≤ 5 px and angle ≤ 1.0°`

Translation

`dx > 5 px or dy > 5 px`

Rotation

`angle > 1.0°`

Missing Label

`no box or no red label detected`

Strengths

- Deterministic and explainable — every verdict traces to a measured value.
- Real-time: lightweight OpenCV ops, no GPU or training data needed.
- Precise sub-pixel shift and sub-degree tilt measurement.

Limitations

- Assumes a red label on a high-contrast box; new colours need new HSV ranges.
- Fixed thresholds and lighting-sensitive HSV — glare can break detection.
- One label per frame; no text or barcode correctness check.

Challenges

Key environmental and dataset limitations preventing stable production deployment.

Physical & Environmental

ISSUE 01 **Glare & Reflections:** When the bottle surface is shiny, the light can reflect on the label and make the camera detect it incorrectly.

ISSUE 02 **Lighting variation:** when the light is too bright or too dark, the system may not identify the label accurately.

ISSUE 03 **Curved Surfaces:** The bottle is non-flat, so the label may appear slightly distorted in the image. This can affect the rotation measurement. .

Data & Calibration

ISSUE 04 **Small Training Dataset:** A small training dataset can make the system weak when testing new bottles or new conditions.

ISSUE 05 **Tolerance Calibration:** If the tolerance is too strict, good products may be rejected. If it is too loose, defective products may pass.

ISSUE 06 **Real-World Viability:** The system must work reliably in a real factory, where there is noise, movement, lighting changes, and reflections.

Suggested Improvement

Practical enhancements to make the inspection system robust, accurate, and production-ready.

Data & Physical Setup

STEP 01 **More & Varied Data:** collect more images with different lighting conditions, different label positions, and different product angles.

STEP 02 **Optimized Lighting:** Using ring light or diffused light can reduce glare, reflection, and shadows.

STEP 03 **Conveyor Testing:** Execute real-time validation on physical conveyor belts to guarantee the system keeps detecting with production line speeds.

Intelligence & Software

STEP 04 **Deep Learning (CNN):** Timprove defect detection because it learns from many examples instead of depending only on fixed rules.

STEP 05 **Auto-Tuned Tolerances:** the tolerance values can be automatically tuned using data instead of setting them manually.

STEP 06 **OCR & Barcode Check:** This step will allow the system to check not only the label position, but also whether the printed information and barcode are correct.

Huibo lab Comparison

Engineering evaluation comparing standard consumer components against the precision Huibo Lab setup.

Label Alignment System

IMAGE SENSOR & SHUTTER

Standard CMOS rolling shutter camera (phone). High noise, auto-exposure, and auto-focus latency.

OPTICS & LENS SYSTEM

Wide-angle consumer lens. Susceptible to radial barrel distortion and parallax (perspective) errors.

ILLUMINATION & LIGHTING

Ambient room lighting (LED). Vulnerable to shadows, color temperature drifts.

PHYSICAL FIXTURING

Manual workpiece placement. Variable camera height, shifting angles, and background variations.

Huibo Lab Setup

IMAGE SENSOR & SHUTTER

High-performance CCD or CMOS global shutter camera. Manual lockable parameters and zero-latency exposure control.

OPTICS & LENS SYSTEM

Telecentric Lens. Constant magnification regardless of object distance; eliminates optical parallax completely.

ILLUMINATION & LIGHTING

Dedicated synchronized ring light. Strobe-synchronized directly with the camera shutter.

PHYSICAL FIXTURING

Rigid anodized aluminum rails. Precision guides, photo-eye triggers, and high-contrast backing plates.

Thank you!